Radiomics and Deep Learning for Tumor Diagnosis and Prognosis: A Comprehensive Survey

Abstract

The integration of radiomics and deep learning (DL) has revolutionized oncological imaging, transitioning from qualitative visual assessment to quantitative, data-driven decision support. This survey provides a comprehensive analysis of 300 recent studies spanning the spectrum of tumor diagnosis, molecular characterization, and survival prognosis. We systematically categorize methodological advancements into handcrafted radiomics, deep radiomics, and hybrid fusion architectures, evaluating their efficacy across diverse malignancies including glioblastoma, non-small cell lung cancer (NSCLC), and head and neck squamous cell carcinoma. Key themes include the evolution from static feature extraction to dynamic and delta-radiomics for longitudinal monitoring, the emergence of radiogenomics for non-invasive mutation prediction, and the critical role of multimodal fusion integrating imaging, genomics, and clinical data. Furthermore, we address persistent challenges such as segmentation variability, data heterogeneity, model interpretability, and the reproducibility crisis in medical AI. By synthesizing evidence from large-scale benchmarks and multi-center studies, this paper outlines the current state-of-the-art and identifies pivotal directions for future research to facilitate clinical translation.

1. Introduction

The paradigm of precision oncology relies heavily on the accurate characterization of tumor phenotype and genotype to guide treatment strategies. Traditionally, tumor assessment has depended on manual measurements and subjective visual interpretation of medical images, which often fail to capture the underlying biological heterogeneity. Radiomics, defined as the high-throughput extraction of quantitative features from medical images, offers a non-invasive mechanism to decode tumor phenotypes invisible to the naked eye [1]. Concurrently, deep learning, particularly convolutional neural networks (CNNs), has demonstrated superior capability in automated feature learning, bypassing the need for manual engineering [2]. The convergence of these fields has yielded robust frameworks for diagnosing tumors, predicting genetic mutations (radiogenomics), and forecasting patient survival outcomes.

Recent years have witnessed an explosion in methodologies ranging from classical machine learning pipelines utilizing handcrafted features to end-to-end deep learning architectures. Early approaches focused on extracting statistical, shape, and texture descriptors from segmented regions of interest (ROIs) using tools like PyRadiomics [3]. However, limitations in reproducibility due to segmentation variability and imaging protocol heterogeneity have driven the development of stability-aware filtering and harmonization techniques [4][5]. Simultaneously, deep learning has evolved from simple 2D classifiers to complex 3D volumetric networks, attention mechanisms, and transformer-based models capable of capturing long-range spatial dependencies [6][7].

A significant trend in the literature is the shift towards multimodal integration. Purely imaging-based models are increasingly being augmented with genomic data, clinical records, and pathological reports to create holistic prognostic signatures [8][9][10]. This "radiophenomic" approach leverages the complementary nature of different data modalities, where imaging captures spatial heterogeneity and genomics provides molecular context [2][11]. Moreover, the field is moving beyond single-timepoint analysis to dynamic radiomics, which quantifies temporal changes in tumor characteristics during therapy, offering early indicators of treatment response [12][13].

Despite these advancements, substantial challenges remain. The “black-box” nature of deep learning models hinders clinical trust, necessitating explainable AI (XAI) techniques that map latent features to interpretable radiomic signatures [14][15]. Furthermore, the generalizability of models across different institutions and scanner types is often compromised by domain shift, prompting research into domain adaptation, semi-supervised learning, and synthetic data generation [16][17]. This survey synthesizes findings from 300 relevant studies to provide a structured overview of the methodological landscape, evaluate performance across various cancer types, and discuss the critical barriers to clinical deployment.

2. Method

The methodological landscape of radiomics and deep learning for oncology can be broadly categorized into three primary streams: handcrafted radiomics pipelines, deep learning-based feature extraction, and hybrid fusion frameworks. Within these streams, specific innovations address segmentation, feature stability, multimodal integration, and survival modeling.

2.1 Handcrafted Radiomics Pipelines and Feature Engineering

Handcrafted radiomics remains a cornerstone of medical image analysis, offering interpretability and established biological correlations. The standard workflow involves image acquisition, ROI segmentation, feature extraction, selection, and modeling [1][18].

2.1.1 Feature Extraction and Stability Modern radiomics pipelines extract hundreds to thousands of features, including first-order statistics, shape descriptors, and higher-order texture matrices (GLCM, GLRLM, GLSZM) [19][3]. However, the stability of these features is highly sensitive to segmentation variability and imaging parameters. Studies have shown that texture features from Whole Tumor (WT) regions exhibit higher stability ($OCCC \geq 0.95$) compared to enhancing core features, which are more susceptible to segmentation errors [4]. To mitigate this, researchers have developed stability-filtering mechanisms, such as using the Overall Concordance Correlation Coefficient (OCCC) to discard unstable features prior to modeling, which has been shown to improve AUROC for IDH mutation prediction by 13% [4].

Furthermore, the choice of feature complexity impacts predictive performance. Evidence suggests that optimal predictive power often resides in lower-complexity features (morphological, intensity, basic texture) rather than complex filter-based transformations, challenging the assumption that more complex features yield better results [20]. Innovations in feature engineering also include the use of mathematical morphology to extract granularity and covariance features, which provide independent prognostic information distinct from classical radiomics in lung cancer [21]. Additionally, dynamic radiomics has emerged as a powerful extension, mapping static feature vectors across time points to capture temporal heterogeneity, outperforming static models in breast cancer diagnosis [12].

2.1.2 Feature Selection and Dimensionality Reduction Given the high dimensionality of radiomic data relative to sample sizes, robust feature selection is critical. Traditional methods include LASSO, Recursive Feature Elimination (RFE), and Principal Component Analysis (PCA) [22][23]. Novel approaches include Gradient-Loss Recursive Feature Elimination (GL-RFE), which uses backpropagation gradients to quantify feature influence, achieving a 1.9% accuracy improvement over conventional RFE in lung cancer staging [24]. Another innovative strategy involves patient-specific feature selection, where a neural network dynamically weights radiomic features based on individual image context, creating a "radiomic fingerprint" that adapts to pathological variations [25].

2.2 Deep Learning Architectures for Radiomic Analysis

Deep learning automates feature extraction, learning hierarchical representations directly from raw image data. Architectures have evolved from standard CNNs to sophisticated 3D networks, transformers, and graph neural networks.

2.2.1 Convolutional Neural Networks (CNNs) and 3D Volumetric Learning 3D CNNs are essential for capturing volumetric context in CT and MRI data. Architectures like 3D-ResNet and EfficientNet have been successfully applied to NSCLC survival analysis, where 3D volumetric features combined with clinical data improved the C-index by 4% over single-modality baselines [8]. In head and neck cancer, dual-branch 3D CNNs processing PET and CT simultaneously have demonstrated superior performance in predicting progression-free survival compared to single-modality models [26].

Innovations in CNN architecture include the integration of attention mechanisms. Convolutional Block Attention Modules (CBAM) have been employed to refine spatial and channel-wise feature maps in 3D radiomic survival networks, although some studies note that fusing noisy clinical data can sometimes degrade performance compared to imaging-only modalities [27]. Furthermore, specialized modules like Octave Convolution (OctConv) separate high and low-frequency features to reduce redundancy, while skip-scSE blocks enhance feature excitation in glioblastoma segmentation and survival prediction [28].

2.2.2 Vision Transformers and Hybrid Architectures Vision Transformers (ViT) have gained traction for their ability to model long-range dependencies. In head and neck cancer, ViT-based models (e.g., UNETR) have achieved segmentation parity with leading 3D CNNs while enabling efficient 2D "Super Image" representations for prognosis [6]. For breast tumor segmentation, Large Kernel MedNeXt models initialized with UpKern algorithms have shown that expanding kernel sizes via weight interpolation improves Dice scores by capturing broader contextual information [29].

Hybrid architectures combining CNNs and Transformers are also prevalent. For instance, radiomics-guided ViTs utilize contrastive loss to align pixel-based embeddings with handcrafted radiomic descriptors, enabling effective token pruning and improving interpretability in survival analysis [30]. Similarly, Swin Transformers have been integrated into multimodal frameworks to process volumetric features alongside clinical variables for distant metastasis prediction, outperforming CT-only baselines [7].

2.2.3 Graph Neural Networks (GNNs) and Multiple Instance Learning (MIL) Graph Neural Networks model the spatial relationships between tumor sub-regions or patches. In computational pathology, WSIs are often treated as graphs where patches are nodes connected by spatial adjacency. Patch-GCN, which models WSIs as 2D point clouds, has outperformed prior weakly supervised approaches in survival prediction across multiple cancer types [31]. For multifocal diseases, injective multiple instance pooling functions (e.g., sum pooling) have been proposed to aggregate risks from multiple lesions, addressing the non-injectivity of standard max/mean pooling and improving AUC in metastatic cancer outcome prediction [32].

In radiogenomics, patient hypergraph networks (PHGN) connect individuals based on latent feature similarity, although linear models sometimes outperform graph approaches when sample sizes are limited [33]. Conversely, heterogeneous graph networks with convolutional masked autoencoders (SELECTOR) have demonstrated robustness in handling missing modalities by reconstructing missing structural semantics via meta-paths [34].

2.3 Hybrid Radiomics-Deep Learning Fusion

Hybrid frameworks combine the interpretability of handcrafted radiomics with the representational power of deep learning. These approaches often mitigate the limitations of small datasets by leveraging pre-trained CNNs for feature extraction while retaining explicit radiomic signatures.

2.3.1 Feature-Level Fusion A common strategy involves concatenating handcrafted radiomic features with deep latent features. In pancreatic cancer early detection, minimizing mutual information between handcrafted and deep radiomic features during Variational Autoencoder (VAE) training forces the deep features to capture orthogonal information, consistently improving AUC [35]. Similarly, in glioblastoma prognosis, fusing handcrafted radiomics with deep latent variables via dissimilarity-based dimension reduction has shown a 10.7% accuracy improvement over baseline models [36].

Deep radiomic signatures (DRFs) represent another fusion paradigm where activation maps from CNN layers are encoded using standard radiomic texture functions (e.g., GLCM, entropy). This approach has been successful in predicting glioma survival by integrating deep texture features with immune cell markers, capturing tissue heterogeneity more effectively than raw activation maps [37][38].

2.3.2 Decision-Level Fusion and Ensemble Methods Decision-level fusion aggregates predictions from separate radiomics and deep learning models. In breast cancer diagnosis, ensembles of ConvNeXt and handcrafted radiomics models, calibrated via histogram matching, have achieved higher AUCs than either modality alone, compensating for individual model weaknesses [39]. For IPMN classification, weighted averaging of sparse deep features and dense radiomics vectors has improved accuracy by 20.6% over ViT baselines in multi-center settings [40].

Ensemble strategies also extend to multi-task learning. DeepMTS frameworks jointly optimize segmentation and survival tasks, where the segmentation backbone guides feature extraction towards tumor regions, implicitly improving prognostic performance [41][42]. However, equal weighting of task losses can be suboptimal, suggesting a need for adaptive loss weighting strategies [41].

2.4 Multimodal Integration: Radiogenomics and Clinical Data

The integration of imaging with genomic, pathological, and clinical data represents the frontier of precision oncology.

2.4.1 Radiogenomics Radiogenomics aims to predict genetic mutations and molecular subtypes non-invasively. In glioblastoma, hybrid models fusing MRI radiomics with deep learning have achieved high accuracy in predicting MGMT methylation and IDH mutation status, although some rigorous benchmarks suggest that standard architectures may struggle to learn these correlations without explicit guidance [43][44]. For lung cancer, genotype-guided radiomics (GGR) employs a two-stage framework where DNN regressors map hybrid CT features to gene expression values, bridging the performance gap with actual genomic data [45].

Advanced radiogenomic pipelines utilize spatial priors and biopsy co-localization. The SpACe framework integrates context-aware radiomic features from biopsy sites with population-level spatial atlases to generate voxel-wise mutation probability maps, significantly enhancing prediction accuracy for EGFR amplification [46]. Additionally, fractal dimension features extracted from MRI have shown statistical significance in differentiating molecular subtypes of glioblastoma, offering a geometric perspective on tumor heterogeneity [47].

2.4.2 Integration with Clinical and Pathological Data Multimodal fusion often incorporates clinical variables (age, stage, biomarkers) and pathological reports. In renal cell carcinoma, selective inclusion of high-importance clinical variables combined with 3D radiomic features significantly outperforms indiscriminate variable fusion [9]. For pancreatic cancer, integrating prediagnostic radiology text embeddings (via Sentence-BERT) with volumetric radiomic features has outperformed unimodal approaches in opportunistic screening [48].

In computational pathology, multimodal models integrate Whole Slide Images (WSIs) with genomic data. Frameworks like PS3 utilize prototype-based fusion to align diagnostic prototypes from pathology reports, histological prototypes from WSIs, and pathway prototypes from RNA-seq, achieving superior C-indices across TCGA cohorts [49]. Hierarchical fusion models (HFGPI) explicitly model the biological cascade from genes to proteins to phenotypes, outperforming flat integration strategies [50].

2.5 Survival Analysis and Prognostic Modeling

Survival analysis in radiomics and deep learning moves beyond classification to time-to-event prediction, handling censored data and competing risks.

2.5.1 Deep Survival Networks Deep learning extensions of the Cox Proportional Hazards model, such as DeepSurv and DeepHit, are widely used. Penalized Deep Partially Linear Cox Models integrate SCAD penalties with DNNs to perform simultaneous feature selection and nonlinear modeling, outperforming standard DeepSurv in lung cancer prognosis [51]. For competing risks, Deep-CR MTLR extends neural MTLR frameworks to handle multiple event types, preventing bias from conflating cancer-specific death with other causes [52].

Innovations include mini-batched partial likelihood formulations that enable scalable training of deep survival models on large datasets by restricting risk set calculations to current batches, mitigating non-convexity issues [53]. Additionally, vector quantization-based models (RobSurv) generate discrete codebook tokens to create noise-invariant anatomical representations, stabilizing cross-modal fusion in survival prediction [54].

2.5.2 Longitudinal and Delta Radiomics Longitudinal analysis captures tumor evolution. Delta radiomics, which quantifies changes in features between time points, has proven effective in distinguishing radiation necrosis from tumor progression and predicting treatment response [55][56]. Frameworks like RAMAC use image registration to establish lesion correspondence across time points, enabling consistent extraction of delta radiomic ratios that improve C-indices in metastatic breast cancer [57][58]. Dynamic radiomics further extends this by modeling feature trajectories via parametric curve fitting, capturing temporal heterogeneity better than static values [12].

2.6 Segmentation and Region of Interest Definition

Accurate segmentation is a prerequisite for radiomics. While manual segmentation is the gold standard, it is labor-intensive and subject to inter-observer variability.

2.6.1 Automated Segmentation Networks nnU-Net remains a dominant architecture for automatic segmentation, often serving as the backbone for radiomics pipelines [59][60]. Enhancements include the integration of Squeeze-and-Excitation (SE) modules to improve PET/CT segmentation accuracy [61] and the use of pseudo-labeling to refine segmentation on multi-center test data [59]. For 3D volumetric segmentation with sparse annotations, SAMONAI extends the Segment Anything Model (SAM) via prompt propagation, enabling robust segmentation from single-point prompts [62][63].

2.6.2 Segmentation-Free and Weakly Supervised Approaches To bypass segmentation bottlenecks, segmentation-free approaches utilize Class Activation Maps (CAMs) or Multi-Angle Maximum Intensity Projections (MA-MIPs). In head and neck cancer, MA-MIPs of PET images combined with object detection cropping have achieved competitive C-indices without explicit tumor delineation [64]. Weakly supervised methods using single-point priors and distance-based loss functions have also shown promise in detecting tumors in PET scans, reducing the reliance on dense masks [65].

2.7 Explainability and Interpretability

Interpretability is crucial for clinical adoption. Techniques range from post-hoc explanations to inherently interpretable architectures.

2.7.1 Mapping Deep Features to Radiomics Methods like Rad4XCNN compute correlations between CNN feature maps and handcrafted radiomic features, mapping opaque deep features to intelligible radiomic signatures (e.g., Energy, Non-Uniformity) [66]. Similarly, clustering-derived feature quantitation has been used to explain up to 80% of Deep Learning System (DLS) score variance in colorectal cancer, identifying specific visual features like Tumor-Adipose Features as independent prognostic markers [67].

2.7.2 Attention and Saliency Maps Attention mechanisms in MIL and Transformers provide inherent interpretability by highlighting influential patches or tokens. In prostate cancer relapse prediction, self-attention weights in MIL models correlate with pathologist-identified regions of interest [68]. Grad-CAM and similar saliency maps are frequently used to visualize regions contributing to predictions, although their clinical utility requires further validation [69].

2.7.3 Concept Bottlenecks and Dictionaries Concept Bottleneck Models (CBMs) predict intermediate clinical concepts (e.g., biomarkers) before the final diagnosis, enhancing transparency. ConRad, a CBM for lung nodules, fuses expert-derived radiomics with DNN-predicted biomarkers, yielding interpretable models that match black-box CNN performance [15]. Additionally, radiomics dictionaries map quantitative features to semantic descriptors (e.g., PI-RADS, Lung-RADS), bridging the gap between computational outputs and clinical reasoning [70][71].

3. Challenges and Outlook

Despite significant progress, the field faces several critical challenges that hinder widespread clinical deployment.

3.1 Data Heterogeneity and Generalizability

A major limitation is the lack of standardized imaging protocols and the resulting domain shift across institutions. Models trained on single-center data often fail to generalize to external cohorts due to variations in scanner types, reconstruction kernels, and acquisition parameters [72][73]. Harmonization techniques like ComBat have shown efficacy in reducing batch effects, but their application to deep learning features requires careful validation [59][74]. Furthermore, small sample sizes remain a pervasive issue, leading to overfitting and unstable feature selection, particularly in rare cancers or specific molecular subtypes [2][75].

3.2 Segmentation Variability and Feature Stability

Radiomic features are highly sensitive to segmentation accuracy. Studies indicate that even moderate segmentation errors (DSC ~0.78) can suffice for survival prediction if edge discrepancies do not alter global texture, but precise contouring remains critical for specific shape features [76]. The variability introduced by different segmentation algorithms or manual delineators can significantly impact model performance, necessitating robustness assessments via simulated perturbations [77][78]. Stability-aware modeling, which filters features based on reproducibility metrics like ICC or OCCC, is becoming a standard practice to ensure reliable biomarkers [4][5].

3.3 Interpretability and Clinical Trust

The "black-box" nature of deep learning models poses a barrier to clinical trust. While XAI techniques like SHAP and Grad-CAM provide insights, they often fail to offer mechanistic explanations or may highlight spurious correlations [79][80]. There is a growing need for inherently interpretable models, such as concept bottlenecks or radiomics dictionaries, that align with clinical reasoning processes [15][70]. Moreover, the disconnect between high-dimensional radiomic features and biological mechanisms requires further investigation to validate imaging biomarkers as surrogates for underlying pathophysiology [81].

3.4 Evaluation and Reproducibility

Reproducibility remains a significant concern. Many studies suffer from methodological flaws, including data leakage, improper cross-validation, and lack of external validation [82][83]. The use of retrospective, single-center datasets limits the generalizability of findings. Initiatives like Openradiomics advocate for standardized protocols and repetitive stochastic evaluation to distinguish stable performance from overfitted outliers [83]. Additionally, the lack of prospective trials means that most models have not been validated in real-world clinical workflows, leaving their actual impact on patient outcomes unknown [84][85].

3.5 Future Directions

Future research should focus on several key areas: 1. **Longitudinal and Dynamic Modeling:** Expanding beyond static snapshots to continuous monitoring of tumor evolution using delta radiomics and recurrent networks will enable earlier detection of treatment resistance [12][13]. 2.

Multimodal Foundation Models: Leveraging large-scale foundation models pre-trained on diverse medical data could improve feature generalizability and reduce the need for task-specific fine-tuning [74][86]. 3. **Causal Inference and Mechanistic Integration:** Integrating mechanistic models (e.g., reaction-diffusion equations) with data-driven AI could provide biologically grounded predictions and improve robustness [87][88]. 4. **Federated Learning and Privacy:** To address data silos and privacy concerns, federated learning frameworks will be essential for training robust models on distributed datasets without sharing raw patient data [89]. 5. **Clinical Implementation Studies:** Rigorous prospective trials are needed to evaluate the clinical utility, cost-effectiveness, and impact on decision-making of radiomics and DL tools [90][85].

4. Conclusion

The integration of radiomics and deep learning has fundamentally transformed the landscape of tumor diagnosis and prognosis. By quantifying tumor heterogeneity and integrating multimodal data, these approaches offer unprecedented insights into tumor biology and patient outcomes. From handcrafted radiomics pipelines enhanced by stability filtering to sophisticated deep learning architectures leveraging transformers and graph networks, the field has made remarkable strides in predictive accuracy. Hybrid models that combine the strengths of explicit feature engineering and automated representation learning have emerged as particularly promising solutions.

However, the path to clinical routine is paved with challenges related to data heterogeneity, segmentation variability, interpretability, and reproducibility. Addressing these issues requires a concerted effort towards standardization, rigorous external validation, and the development of inherently interpretable models. As the field matures, the focus must shift from retrospective performance metrics to prospective clinical utility, ensuring that these advanced computational tools translate into tangible benefits for cancer patients. The future of oncological imaging lies in the seamless fusion of quantitative imaging, genomics, and clinical expertise, driven by robust, transparent, and generalizable AI systems.

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